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Psychometric Characteristics of the Romanian Adaptation of the GAD-7

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Abstract

The GAD-7 is a practical and short self-report measure that has been widely utilized for the screening of generalized anxiety disorder (GAD), a severe form of anxiety disorder that is difficult to identify. Previous studies have investigated the GAD-7 in different settings and various cultures, but often without reporting important internal validity-related indices. To determine the psychometric properties of the GAD-7, the present study examined the internal consistency, convergent validity, underlying factor structure and item discrimination using both Classical Test Theory (CTT) and Item-Response Theory (IRT) models, within both a clinical and a non-clinical sample from Romania. A total of 747 participants, of whom 181 were clinical patients and 566 recruited from the general population, completed the Romanian version of the GAD-7, the Depression, Anxiety, and Stress Scale–21 (DASS-21), and the State-Trait Anxiety Inventory (STAI). CFA results confirmed that a modified one-factor model was an acceptable solution for the clinical sample. The internal consistency of the scale was satisfactory for both samples. The GAD-7 strongly correlated with the three subscales of the DASS-21 and exhibited moderate correlations with the STAI, confirming convergent validity. Keywords: GAD-7; Psychometric properties; Adaptation; IRT; DASS-21; STAI

Keywords GAD-7 · Psychometric properties · Adaptation · IRT · DASS-21 · STAI

Introduction

Generalized anxiety disorder (GAD) is widely recognized as a psychiatric disorder characterized by an internal state of persistent and excessive worry about events and activities that one finds hard to control (Patriquin & Mathew, 2017). The diagnostic criteria for GAD include excessive anxiety and worry that are difficult to control, along with other associated symptoms like restlessness, fatigability, concentration difficulties, irritability, muscle tension and sleep

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disturbances, which impair the individual's psycho-social functioning and could not be attributed to substance abuse or the presence of a medical condition (American Psychiatric Association, 2013).

Over time, many scales have been developed to assess anxiety symptoms. A search for anxiety scales revealed the State-Trait Anxiety Inventory (STAI; Spielberger et al., 1983), the Beck Anxiety Inventory (BAI; Beck and Steer, 1993), and the Generalized Anxiety Disorder (GAD-7; Spitzer et al., 2006) to be highly cited and used by researchers and practitioners worldwide. For example, according to the Web of Science citation report, the STAI has been cited over 133.512 times, BAI has 106.831 citations, and the GAD-7 has received over 48.096 citations (see Table 1). In addition to these well-known measurements, the Depression Anxiety Stress Scales (Lovibond & Lovibond, 1995), the Hamilton Anxiety Scale (HAM-A; Hamilton, 1959), and the Hospital Anxiety and Depression Scale (HADS; Zigmond and Snaith, 1983) have gained popularity. Table 1 describes the citations of the scales mentioned above based on these findings from Web of Science.

Table 1 Citation Report on Anxiety Scales

Name of the scale	Abbreviation	Authors	Times cited	Publications	H-Index
Beck anxiety inventory	BAI	Beck and Steer (1993)	57,306	2450	94
Depression, anxiety, stress scale	DASS-21	Lovibond and Lovibond (1995)	33,569	1697	68
Generalized anxiety disorder scale	GAD-7	Spitzer et al. (2006)	48,096	2338	78
Hamilton anxiety scale	HAM-A	Hamilton (1959)	20,010	909	66
Hospital anxiety and depression scale	HADS	Zigmond and Snaith (1983)	11,442	535	54
State-trait anxiety inventory	STAI	Spielberger et al. (1983)	133,512	5837	132

GAD was first introduced as a distinct diagnosis in the Diagnostic and Statistical Manual of Mental Disorders III-R (DSM-III), and later in the DSM-IV and DSM-5 editions (American Psychiatric Association, 1980, 1994, 2013). According to DSM-5 (American Psychiatric Association, 2013), GAD is described as excessive anxiety and worries about several events or activities. Individuals diagnosed with GAD present difficulties controlling such worries and keeping worrisome thoughts from interfering with attention to tasks. To improve the evaluation of this disorder, an assessment tool for investigating the specific symptoms of GAD has been developed, namely the Generalized Anxiety Disorders-7 (GAD-7; Spitzer et al., 2006). The GAD-7 was built for accurate diagnosis, thus properly guiding mental health clinicians in the process of clinical interventions. Since then, GAD-7 (Spitzer et al., 2006) has been widely used to measure anxiety symptoms, averaging a number of approximately 948.71 citations per year according to the Web of Science citation report on GAD-7. The instrument was developed according to the DSM-IV criteria for GAD, in the form of a brief self-report questionnaire designed to measure anxiety symptoms during the past two weeks (Löwe et al., 2008; Spitzer et al., 2006; Williams, 2014).

Numerous authors have investigated the psychometric properties of GAD-7 in diverse settings (e.g., clinical, non-clinical, general population, epilepsy) and countries (see Table 2). For example, findings from one study (Jordan et al., 2017) revealed that using the GAD-7 diminishes the time required for initiating a psychological treatment, before using additional services addressing somatic symptoms, which suggests the feasibility of this measure for the screening of GAD symptoms. According to Sousa et al. (2015) reports good psychometric characteristics for the Portuguese version of the GAD-7. Good internal consistency and validity were found in numerous other studies (e.g., Johnson et al., 2019; Löwe et al., 2008). In light of these results, the cutoff point of 8 was considered a threshold value for cases in which further evaluation related to the strong expression of anxiety symptoms would be recommended (Johnson et al., 2019; Kroenke et al., 2007). Adjacent to these data, a study in Finland established a cutoff point of 7 for Finland's clinical population (Kujanpää et al., 2014).

Due to its brief structure and ease of use, the GAD-7 has inadvertently attracted the attention of researchers interested in adapting the scale to their native language. To estimate the number of adaptations, we have searched the Web of Science database for all articles that were adaptations of the GAD-7. Table 2 shows the results of our systematic search, including the reliabilities and model fit indices obtained for each research paper. Overall, the reliability indices represented by Cronbach's α ranged from 0.82 to 0.95 suggesting good internal consistency across adaptations. In addition,

Table 2 Adaptation of the GAD-7 across Countries

Authors	Country	*N*	Cronbach's α	CFI	TLI	RMSEA	SRMR
Ahn et al. (2019)	Korea	1157	0.93	-	-	-	-
Franco-Jimenez and Nunez-Magallanes (2022)	Peru	407	0.89	0.99	0.99	0.06	0.03
De Man et al. (2021)	India	1209	-	0.94	0.90	0.11	0.04
Dhira et al. (2021)	Bangladesh	677	0.90	0.98	0.98	0.07	0.03
Donker et al. (2011)	Netherlands	502	0.92				
Garcia-Campayo et al. (2010)	Spain	212	0.94	0.99	-	0.08	-
Hinz et al. (2017)	Germany	9721	0.85	0.97	0.96	0.09	-
Konkan et al. (2013)	Turkey	222	0.85	0.99	0.99	-	-
Nyongesa et al. (2020)	Kenya	450	0.82	1.00	1.00	0.01	-
Sawaya et al. (2016)	Lebanon	186	0.95	-	-	-	-
Sousa et al. (2015)	Portugal	100	0.88	-	-	-	-
Tong et al. (2016)	China	213	0.89	-	-	-	-
Woon et al. (2020)	Malaysia	300	0.91	0.95	0.93	0.12	-
Zinchuk et al. (2021)	Russia	233	0.91	-	-	-	-

one study (De Man et al., 2021) reported total omega (0.80) and hierarchical omega (0.72), which supports the reliability of the GAD-7.

Eight studies have conducted a confirmatory factor analysis (CFA) and have reported model fit indices for the one-factor solution of the GAD-7. The comparative fit index (CFI), the Tucker Lewis Index (TLI), the root mean square error of approximation (RMSEA), and the standardized root mean square residual (SRMR) are commonly used fit indexes reported when describing the fit of structural equation models in CFAs. The CFI and TLI are both incremental fit indexes that assess the improvement in the fit of a model over that of a baseline model with no relationship among the model variables. Larger values (>0.90) indicate better model fit (Kline, 2010). The RMSEA provides information about 'badness of fit', with lower RMSEA values indicating good model fit (Kline, 2010). Finally, the SRMR measures the mean absolute correlation residual, with smaller values suggesting a good model fit (Kline, 2010).

As a result, we have included the four fit indexes (i.e., CFI, TLI, RMSEA, and SRMS) for the one-factor solutions reported by the research papers included in our systematic search. Overall, the eight articles that reported the fit indexes revealed acceptable incremental fit indexes (CFI values ranged from 0.94 to 1.00, and TLI values ranged from 0.90 to 1.00). Only seven studies have reported the RMSEA with scores ranging from 0.01 to 0.12. Of these seven studies, only one has obtained an acceptable RMSEA value (Nyongesa et al., 2020), while the others exceeded the threshold of 0.05. Only three studies have reported SRMS with acceptable scores that ranged from 0.03 to 0.04. However, six studies did not conduct SEM analysis of the CFAs.

Since mental health specialists in Romania are confronted with a wide range of psychiatric symptoms in patients admitted to both inpatient and outpatient units providing care for their somatic illness and psychiatric illnesses, we consider that the utility of using GAD-7 resides in having an accurate assessment of GAD in accordance to its psychometric properties in the Romanian population. This may also help with the assessment of the disorder severity, which increases the chances of an appropriate psychological and medical treatment of patients suffering from GAD.

This study has two primary objectives. Firstly, it seeks to examine and evaluate the psychometric properties of the GAD-7 scale within the context of both clinical and nonclinical population samples in Romania. By conducting a rigorous analysis, the study focuses on the reliability, validity, and factor structure of the GAD-7 when applied to individuals in different settings and with varying anxiety levels. Secondly, the study endeavors to address a practical need by developing and providing practitioners and health specialists in Romania with an empirically-supported test adaptation of the GAD-7. The intention behind this objective is to equip professionals with a validated instrument that accurately captures and measures anxiety symptoms, thereby facilitating accurate diagnosis, monitoring, and treatment planning for individuals in Romania.

Method

Participants and Procedure

A total number of 747 participants were included in the study. Of these, 566 participants (75.76%) were recruited from the general population, and 181 were clinical patients with or without a psychiatric diagnosis. Clinical patients were recruited from 4 large hospitals in 4 different cities in Romania: Bucharest (the University Emergency Hospital Bucharest), Iasi (the Psychiatric Institute "Socola" Iaşi), Satu Mare (the County Emergency Hospital of Satu Mare), and Targu Mures (the County Emergency Clinical Hospital of Târgu-Mureş).

All the data were collected using the same procedure for all clinical units involved in the study. Thus, a single sheet of paper with the whole set of 48 items (i.e., from all four psychological instruments employed) was administered to participants in paper-and-pencil format under the supervision of a clinical psychologist. All the items were rated on a four-step Likert scale. In addition, socio-demographic data (e.g., sex, marital status, occupation, years of education, place or origin, personal income) were also collected. Exclusion criteria were: cancer palliative care, dementia or mental retardation, and the inability to understand and/or write the informed consent. Patients admitted to inpatient and outpatient psychiatric units without a previous psychiatric diagnosis and patients who did not take medication three months before the start of the study were included in the study.

The original authors of the GAD-7 (Spitzer et al., 2006) have given written consent for using the GAD-7 in clinical research settings for the Romanian population, and have provided us with the authorized Romanian version of the scale (i.e., the translation with a Linguistic Validation Certificate from MAPI Research Institute). The study was approved by the Ethical Committee of the University Emergency Hospital, Bucharest as well as by the County Emergency Clinical Hospital of Târgu-Mureş, and County Emergency Hospital of Satu Mare.

Measures

The Generalized Anxiety Disorders Scale-7 measures anxiety symptoms in different kinds of settings and populations (Spitzer et al., 2006). The GAD-7 is composed of 7 items rated on a 4-point Likert scale (0 – "Not at all" to 3 – "Nearly every day"). Example items are "Feeling nervous, anxious or on edge", "Not being able to stop or control worrying", and "Feeling afraid as if something awful might happen". Scores range between 0 and 21, with the cutoff points of 5, 10 and 15 representing mild, moderate and severe anxiety. It is generally agreed (Johnson et al., 2019) that for scores placed over a cutoff point of 10, further evaluation is needed in order to establish a final diagnosis.

The State-Trait Anxiety Inventory (STAI- Romanian Version) is a psychological questionnaire that measures state and trait anxiety in clinical settings or in the general population (Spielberger et al., 1983). Example items are "I am tense", "I am worried", "I feel calm". All the items are rated on a 4-point Likert scale. The current study only administered the 20 items of the trait-anxiety scale. The STAI trait-anxiety scale had an internal consistency of 0.79 in the present study.

The Depression Anxiety Stress Scale (DASS-21 Romanian Version) is a self-report psychological instrument consisting of 21 items designed to measure depression, anxiety and stress (Lovibond & Lovibond, 1995). Each DASS scale contains seven items. It measures symptoms of depression (e.g., "I couldn't seem to experience any positive feeling at all", "I found it difficult to work up the initiative to do things"), anxiety (e.g., "I felt I was close to panic", "I found myself getting agitated"), and general stress (e.g., "I found it hard to wind down", "I tended to over-react to situations"). All items were rated on a 4-point Likert scale from 0 ("did not apply to me") to 3 ("applied to me very much or most of the time"), with a requirement for participants to report their experience "*over the preceding two weeks*". The DASS-21 had an internal consistency of 0.94 for depression, 0.90 for anxiety, and 0.93 for stress in the present study. The instrument is used mainly in clinical settings to screen patients with emotional disturbances.

Results

Descriptive Analyses

The general population and clinical samples were divided into five age groups (see Table 3). Age and gender differences were tested with *t*-tests and ANOVA, respectively. Results revealed no gender differences between males and females (M_{males} = 7.87; $M_{females}$ = 7,28; $t(136) = 0.70$, $p = .48$) and no age differences between the age groups on the GAD-7 scores ($F[4, 172] = 0.30$, $p = .87$) in the clinical sample. Similarly, the general population sample did not show gender differences ($M_{males} = 3.72$; $M_{females} = 4.28$; $t(542) = 1.79$, $p = .07$) and no age differences on the GAD-7 scores ($F[4, 561] = 1.57$, $p = .18$).

Reliability and Factor Structure of the GAD-7

We first investigated the psychometric properties of the GAD-7 on the clinical sample. The GAD-7 items were highly consistent (Cronbach's $\alpha = 0.92$). The item-total correlations were high for all items (0.71 −0.85). Bartlett's test was significant, $\chi^2(21) = 856.93$, $p < .001$, and the KMO measure of sampling adequacy was overall high (0.89), indicating a good relationship between the data and an adequate sample for factor analysis. Inspection of the scree plot and parallel analysis support a one-factor solution. The eigenvalue of the first component was 4.42, and the items

Table 3 Descriptive Statistics

		Male			Female			Total		
	Age groups	*n*	%	Cum. *n*	*n*	%	Cum. *n*	*n*	%	Cum. *n*
CS										
	14–18	2	3%	2	3	3%	3	5	3%	5
	18–25	6	10%	8	11	9%	14	17	10%	22
	25–40	12	21%	20	38	32%	52	50	28%	72
	40–65	32	55%	52	63	53%	115	95	54%	167
	65+	6	10%	58	4	3%	119	10	6%	177
GPS										
	14–18	12	4%	12	6	2%	6	18	3%	18
	18–25	33	12%	45	29	10%	35	62	10%	80
	25–40	80	28%	125	85	30%	120	165	29%	245
	40–65	134	47%	259	143	51%	263	277	48%	522
	65+	25	8%	284	19	7%	282	44	7%	566

Note. CS – clinical sample; GPS – general population sample

Table 4 Factor Loadings for the GAD-7

	Clinical sample			General population	
	EFA	CFA	Modified CFA	EFA	CFA
Item 1	0.80	0.80	0.83	0.55	0.55
Item 2	0.77	0.77	0.75	0.58	0.58
Item 3	0.78	0.77	0.81	0.49	0.49
Item 4	0.75	0.76	0.75	0.56	0.56
Item 5	0.89	0.89	0.95	0.66	0.66
Item 6	0.80	0.81	0.78	0.50	0.50
Item 7	0.75	0.76	0.74	0.51	0.51

Note. All factor loadings are standardized values

explained 63% of the total variance. The factor loadings of all items ranged from 0.75 to 0.89 and model fit measures were adequate ($\chi^2[14] = 63.98$, $p < .001$; TLI = 0.91, BIC = -8.73, RMSEA = 0.14). While the CFI, GFI, and SRMR were acceptable, the RMSEA was not.

Based on the exploratory factor analysis, a one-factor structure was specified for the CFA. Confirmatory factor analysis revealed acceptable model fit indices except for RMSEA ($\chi^2[14] = 64.82$, $p < .001$; CFI = 0.94, TLI = 0.91, RMSEA = 0.14, SRMR = 0.04). We have further inspected modification indices to address the misfit for the one-factor model. Examination of the modification indices recommended allowing the error terms of items one ("Feeling nervous, anxious or on edge"), five ("Being so restless that it is hard to sit still") and seven ("Feeling afraid as if something awful might happen") and items three ("Worrying too much about different things") and five to covary would improve the model ($\chi^2[11] = 20.49$, $p < .05$; CFI = 0.99, TLI = 0.98, RMSEA = 0.07, SRMR = 0.03).

All standardized factor loadings for the one-factor and modified one-factor models were significant at $p < .001$. The factor loadings for the modified one-factor model ranged from 0.74 to 0.95 and were statistically significant at $p < .001$, highlighting good factor loadings (see Table 4).

For the general population sample, the GAD-7 items had a lower internal consistency (Cronbach's $\alpha = 0.75$) compared to the clinical sample. The item-total correlations were high for all items (0.70 −0.73). Bartlett's test was significant, $\chi^2(21) = 678.37$, $p < .001$, and the KMO measure of sampling adequacy was overall high (0.83). The eigenvalue of the first component was 2.13, and the items explained 30% of the total variance. The factor loadings of all items ranged from 0.49 to 0.66 and model fit measures were acceptable ($\chi^2[14] = 36.46$, $p < .001$; TLI = 0.95, BIC = -52.28, RMSEA = 0.05). A one-factor structure was specified for the CFA resulting in satisfactory fit indices ($\chi^2[14] = 36.72$, $p < .001$; CFI = 0.97, TLI = 0.95, RMSEA = 0.05, SRMR = 0.03). As a result, we have not inspected modification indices as no misfit was identified. All standardized factor loadings for the one-factor model ranged from 49 to 0.65 were statistically significant at $p < .001$.

Measurement Invariance

We have tested for measurement invariance by evaluating the differences between the clinical sample and general population sample in terms of configural invariance, metric invariance, scalar invariance, and strict invariance. The results are reported in Table 5 and follow the strict guidelines offered by Putnick and Bornstein (2016). Findings related to configural invariance revealed a good model fit for the configural model. The chi-square difference test was not statistically significant for the metric model ($\Delta\chi^2$ = -10.88, df = 6, $p = .91$) suggesting that factor loadings were equal across groups. Scalar invariance was also observed ($\Delta\chi^2$ = -6.78, df = 6, $p = .33$). Lastly, for strict invariance, we built a model where residuals are constrained to be equal for the clinical and general population samples. The output revealed that the chi-square difference test was significant

Table 5 Measurement Invariance across Clinical and General Population Samples based on CFA

Model	χ2 (df)	CFI	RMSEA	SRMR	Model comparison	Δχ2 (df)	ΔCFI	ΔRMSEA	ΔSRMR	Decision
M1: configural invariance	49.97^{**} (28)	0.98	0.05	0.03		-	-	-	-	-
M2: metric invariance	39.09 (34)	0.99	0.02	0.03	M1	-10.88 (6)	0.01	−0.03	0.00	Accept
M3: scalar invariance	45.87 (40)	0.99	0.02	0.03	M2	6.78 (6)	0.00	0.00	0.00	Accept
M4: strict invariance	72.85 (47)	0.97	0.04	0.05	M3	26.98^{***} (7)	−0.02	0.02	0.02	Reject

Note. $N = 747$; clinical sample $n = 181$; general population sample $n = 566$

$^{**}p < .01$; $^{***}p < .001$

Table 6 Correlations for the Variables in the Study for the Clinical and General Population Samples

		M	*SD*	1	2	3	4	5
1	GAD-7	7.64	5.68	(0.92)/(0.75)	0.68	0.65	0.68	0.42
2	DASS - D	5.60	6.06	0.82	(0.94)/(0.80)	0.71	0.71	0.47
3	DASS - A	4.85	5.12	0.79	0.84	(0.90)/(0.75)	0.75	0.44
4	DASS - S	6.20	5.72	0.86	0.87	0.88	(0.93)/(0.80)	0.44
5	STAI	25.17	8.97	0.57	0.62	0.59	0.60	(0.79)/(0.80)

Note. The values below the diagonal are the correlation coefficients for the clinical sample and values on the diagonal in parentheses are the internal consistencies for each scale. Values above the diagonal are correlation coefficients for the general population sample and values on the diagonal in parentheses are the internal consistencies for each scale

Table 7 Polytomous IRT Coefficients for the GAD-7 Items

	Extremity parameter 1		Extremity parameter 2		Extremity parameter 3		Discrimination	
	CS	GPS	CS	GPS	CS	GPS	CS	GPS
Item 1	-0.92	0.37	0.61	1.53	1.24	2.58	3.05	1.38
Item 2	-0.50	0.68	0.84	1.84	1.35	2.78	2.45	1.46
Item 3	-0.81	0.30	0.42	1.76	1.57	3.09	2.73	1.13
Item 4	-0.75	0.40	0.44	1.54	1.33	3.01	2.45	1.37
Item 5	-0.37	0.47	0.50	1.40	1.17	2.47	4.07	1.86
Item 6	-0.58	0.56	0.61	1.96	1.33	3.00	3.37	1.17
Item 7	-0.34	0.72	0.58	1.71	1.24	2.90	2.98	1.20

Note. CS – clinical sample; GPS – general population sample

($\Delta\chi^2 = 26.98$, df = 7, $p < .001$) suggesting non-equivalence in terms of strict invariance. Such findings are expected considering the differences in sizes across the two groups. These results support a conclusion of partial invariance for the GAD-7.

Convergent Validity

The convergent validity of the GAD-7 was determined by computing Pearson correlations with other similar measures (i.e., DASS-21 and STAI). The results for the clinical sample revealed a moderate relationship between GAD-7 and STAI ($r = .57$, $p < .001$), and higher correlations with DASS (Depression, $r = .82$, $p < .001$; Anxiety, $r = .79$, $p < .001$; Stress, $r = .86$, $p < .001$) (see Table 6). For the general population sample, results revealed a moderate relation between GAD-7 and STAI ($r = .42$, $p < .001$), and slightly higher correlations with DASS (Depression, $r = .68$, $p < .001$; Anxiety, $r = .65$, $p < .001$; Stress, $r = .68$, $p < .001$).

Item Discrimination

Table 7 shows values ordered based on the difficulty level for each item in the clinical sample. For example, item five showed an increase in discrimination, suggesting that this item is helpful to differentiate well between participants with low and high levels of generalized anxiety that endorse the GAD-7. For the general population sample, results revealed no item to sufficiently discriminate by itself between high vs. low levels of generalized anxiety.

Item Response Curves

Figure 1 incorporates the item response category curves for the GAD-7 items. For scales to function appropriately, it is required that responses capture both low and high levels of the construct. Results showed that participants in the clinical sample used the 4-point frequency scale as expected. In addition, the categories displayed no disordering, further suggesting that the response format based on four values provides a proper categorization for measuring anxiety symptoms.

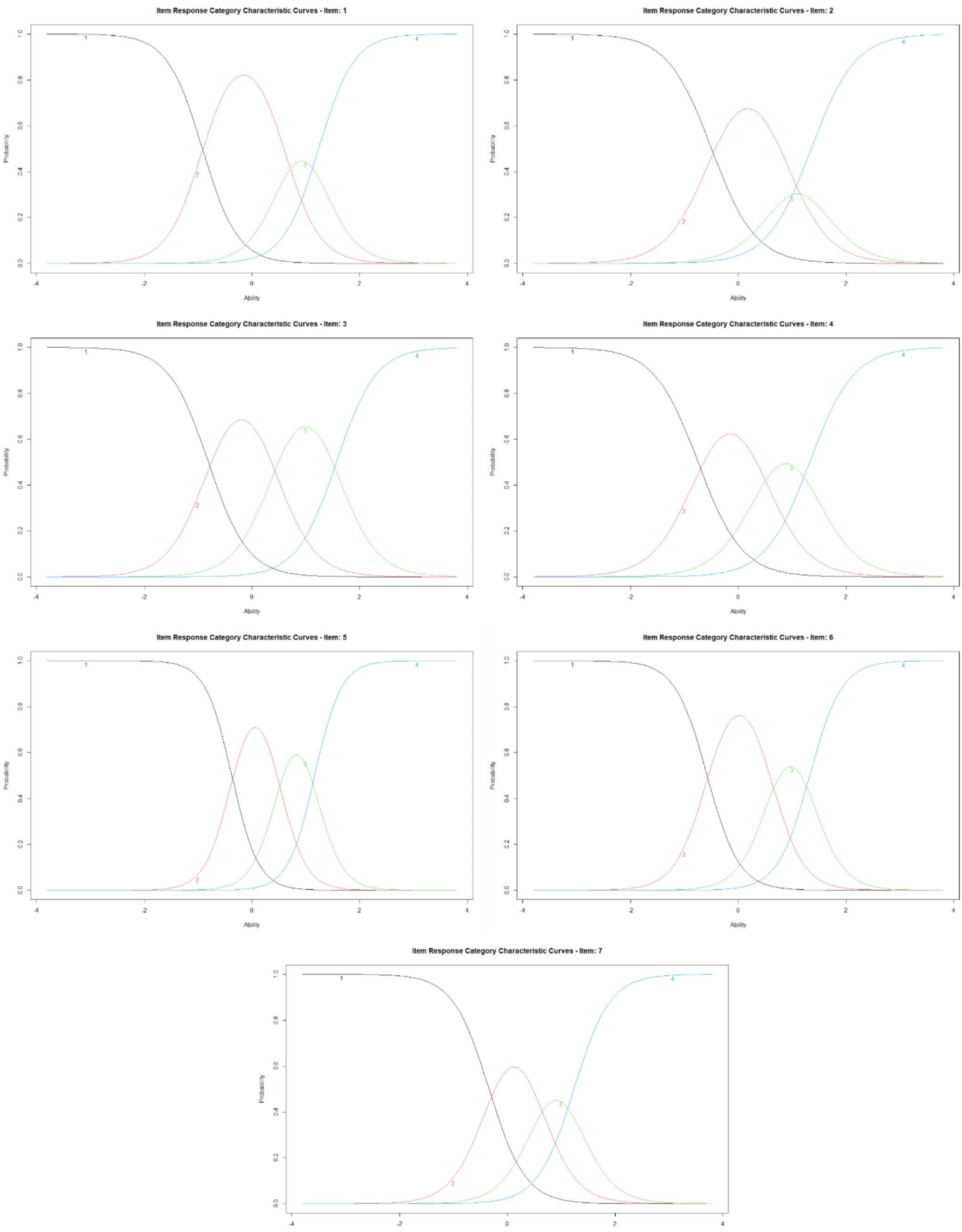


Fig. 1 Item Response Curves for the GAD-7 Items

Item Information Curves and Test Information Function

Item information curves show how much information about anxiety is related to an item. The curves are displayed in Fig. 2. As shown, all seven items of the GAD provide sufficient information about anxiety in the clinical sample. Furthermore, the curve peaks are close to neutral difficulty levels, suggesting that no very "easy" or "difficult" items hinder participants when endorsing the GAD-7. Additionally, as previously stated, item five shows higher levels of discrimination, suggesting that this item differentiates very well between participants with low and high anxiety levels.

The test information function is the sum of all the item information functions. Based on our findings within the clinical sample (see Fig. 2), the test information function shows that the GAD-7 is reasonably informative for most participants with average standing on the latent trait.

Discussion

Summary of Findings

To the best of our knowledge, this is the first study to address the psychometric properties of the Romanian version of GAD-7 for both clinical and non-clinical settings (i.e., the general population). Results confirm the good quality of the GAD-7, from the perspective of internal, as well as external validity criteria.

The GAD-7 scale exhibited satisfactory internal consistency for the clinical sample (Cronbach's $\alpha = 0.92$) and a good internal consistency for the general population sample (Cronbach's $\alpha = 0.75$). Considering both the test information function and item information curves, along with the internal consistency of the GAD-7 on the clinical sample (Cronbach's $\alpha = 0.92$), we conclude that a somewhat higher reliability would be desirable if the GAD-7 were to be classified as a high-stake screening test.

The conditions of convergent validity were met as the Romanian version of the GAD-7 showed significant positive correlations with the DASS scales (i.e., depression, anxiety, and stress) and a moderate positive correlation with the STAI trait anxiety score. Similar results were found for the general population, although the correlation values were overall slightly lower. Results revealed no age and gender differences on the GAD-7 for either the clinical sample or the general population sample.

The unidimensional model used in the CFA analysis for the clinical sample showed a marginal fit to our context. Therefore, the original model was amended using modification indices. Dependency between the error terms of item

Fig. 2 Item Information Curves and Test Information Function for the GAD-7

one ("Feeling nervous, anxious or on edge"), five ("Being so restless that it is hard to sit still") and seven ("Feeling afraid as if something awful might happen"), and items three ("Worrying too much about different things") and five upgraded the fitness of the model. Similar findings were observed in a study conducted on a sample of university students in Bangladesh (Dhira et al., 2021). The unidimensional model for the general population sample showed good fit indices; therefore, a modified one-factor solution was not needed.

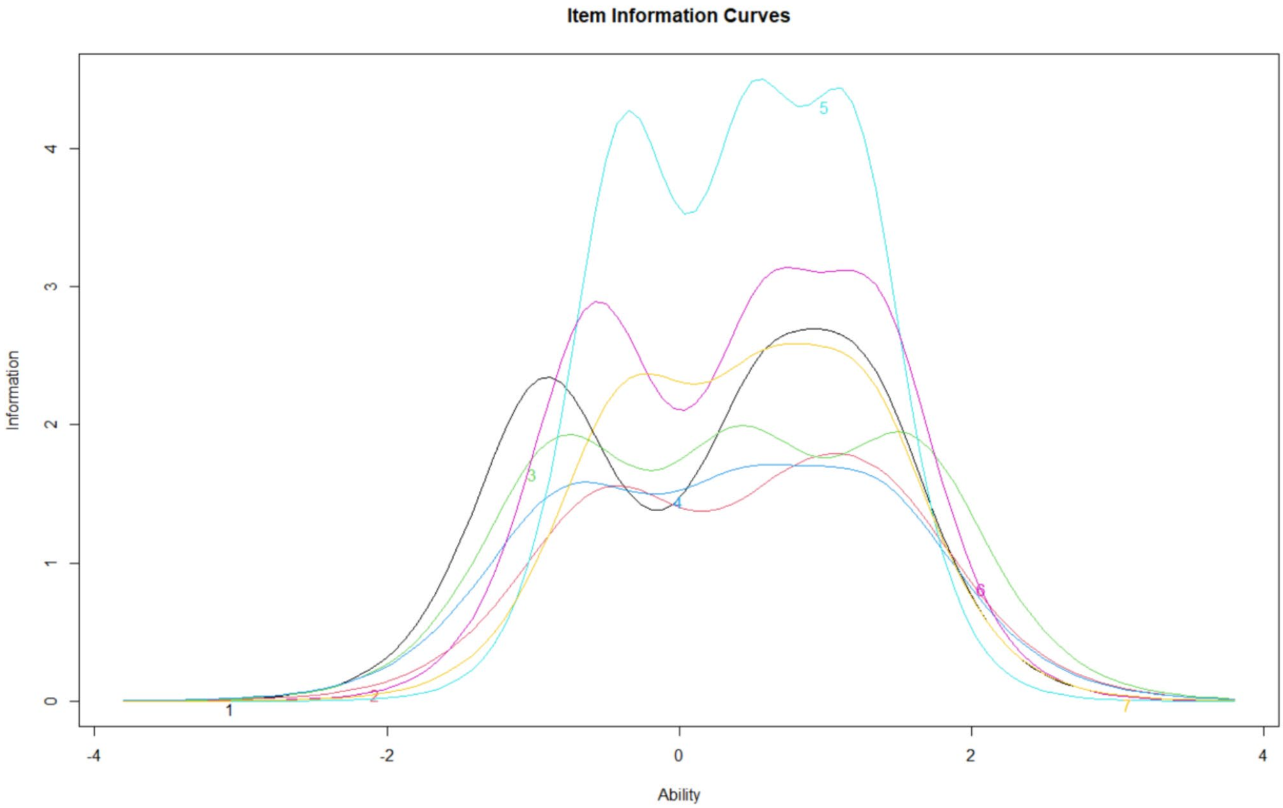
Item discrimination analysis revealed that item five from the GAD-7 differentiates very well between individuals with low and high levels of generalized anxiety in the clinical sample. Conversely, no items were identified to sufficiently discriminate among high- and low-scoring participants from the general population.

Limitations

Although the present study included participants from clinical in conjunction with non-clinical settings (i.e., the general population), several limitations need to be acknowledged while interpreting the results of this study. First, the GAD-7 has shown to be correlated with other related health outcomes indices such as the HADS (Kertz et al., 2013). Therefore, future studies could also include other related measures for a more robust test for the convergent validity of the GAD-7. Second, we did not assess the test-retest reliability of the GAD-7; future studies could explore which item scores remain stable over time. Third, participants from the general population were recruited through convenience sampling, which has several drawbacks, including impediments regarding the results generalizability.

Conclusions

Overall, the GAD-7 scale is an easy-to-score, self-report measure of core generalized anxiety symptoms. It can be used efficiently to assess and identify those patients who might be diagnosed with GAD and/or require clinical, psychological, or psychiatric intervention. Furthermore, the GAD-7 can be easily administered before and after interventions to assess whether improvements were observed or not. The present findings also support the growing evidence of the GAD-7 as a concise and simply administered self-reported questionnaire while also enabling practitioners in Romania to diagnose and treat patients suffering from GAD.


Item Information Curves
Information
Ability
1
2
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-4
-2
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0
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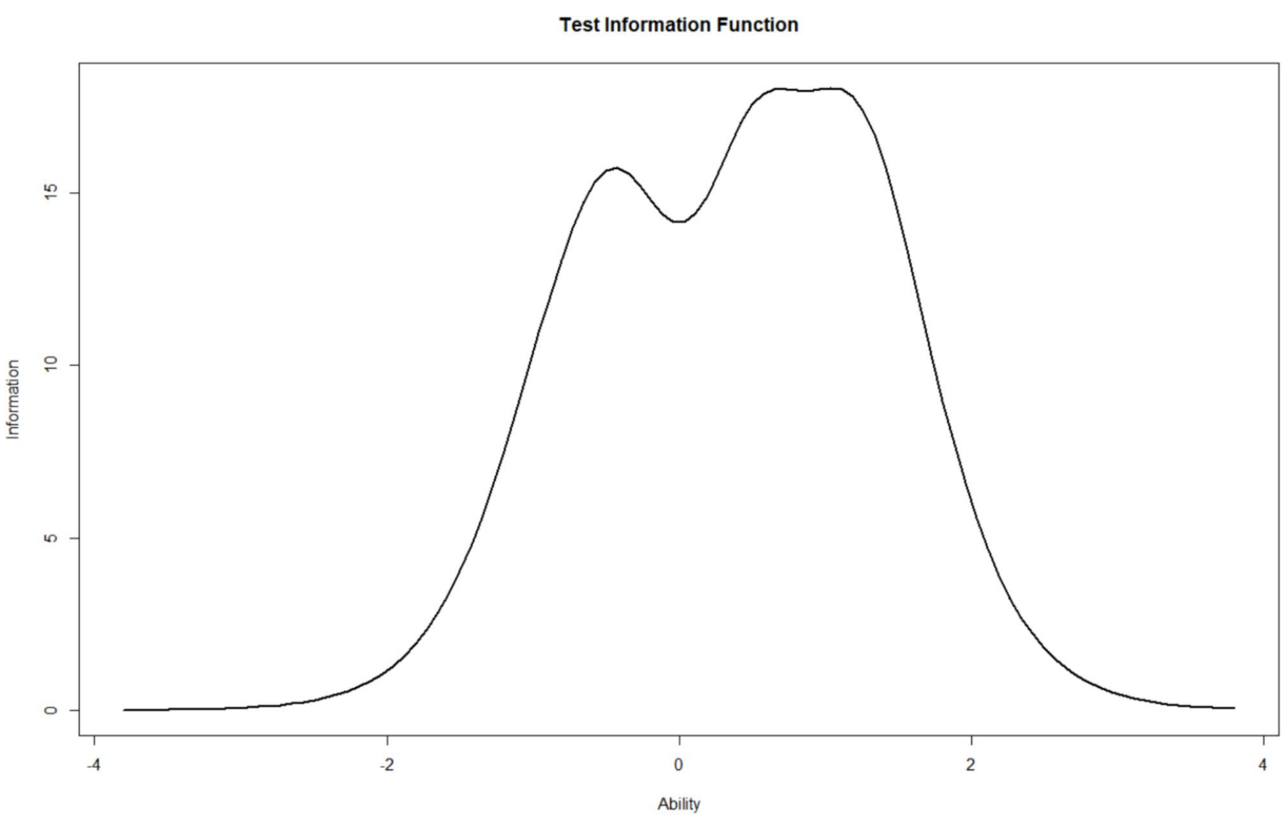
Test Information Function
Information
Ability
-4
-2
0
2
4
0
5
10
15

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Availability of Data and Materials Data used for the present study can be sought upon request.

Declarations

Conflicts of Interest There are no conflicts of interest.

Ethics Approval The original authors of the GAD-7 (Spitzer et al., 2006) have given written consent for using the GAD-7 in clinical research settings for the Romanian population, and have provided us with the authorized Romanian version of the scale (i.e., the translation with a Linguistic Validation Certificate from MAPI Research Institute). The study was approved by the Ethical Committee of the University Emergency Hospital, Bucharest as well as by the County Emergency Clinical Hospital of T?rgu-Mure?, and County Emergency Hospital of Satu Mare.

References

Ahn, J. K., Kim, Y., & Choi, K. H. (2019). The Psychometric Properties and Clinical Utility of the Korean Version of GAD-7 and GAD-2. *Frontiers in Psychiatry*, *10*, 127. https://doi.org/10.3389/fpsyt.2019.00127.

American Psychiatric Association. (1980). *Diagnostic and statistical manual of mental disorders*. III-R).

American Psychiatric Association. (1994). *Diagnostic and statistical manual of mental disorders*. IV).

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders*. V).

Beck, A. T., & Steer, R. A. (1993). *Beck anxiety Inventory Manual*. Psychological Corporation.

De Man, J., Absetz, P., Sathish, T., Desloge, A., Haregu, T., Oldenburg, B., Johnson, L. C. M., Thankappan, K. R., & Williams, E. D. (2021). Are the PHQ-9 and GAD-7 suitable for Use in India? A psychometric analysis. *Frontiers in Psychology*, *12*, 676398. https://doi.org/10.3389/fpsyg.2021.676398.

Dhira, T. A., Rahman, M. A., Sarker, A. R., & Mehareen, J. (2021). Validity and reliability of the generalized anxiety Disorder-7 (GAD-7) among university students of Bangladesh. *Plos One*, *16*(12), e0261590. https://doi.org/10.1371/journal.pone.0261590.

Donker, T., van Straten, A., Marks, I., & Cuijpers, P. (2011). Quick and easy self-rating of generalized anxiety disorder: Validity of the dutch web-based GAD-7, GAD-2 and GAD-SI. *Psychiatry Research*, *188*(1), 58–64. https://doi.org/10.1016/j.psychres.2011.01.016.

Franco-Jimenez, R. A., & Nunez-Magallanes, A. (2022). Psychometric Properties of the GAD-7, GAD-2, and GAD-Mini in peruvian College students. *Propositos Y Representaciones*, *10*(1), e1437. https://doi.org/10.20511/pyr2022.v10n1.1437.

Garcia-Campayo, J., Zamorano, E., Ruiz, M. A., Pardo, A., Perez-Paramo, M., Lopez-Gomez, V., Freire, O., & Rejas, J. (2010). Cultural adaptation into Spanish of the generalized anxiety disorder-7 (GAD-7) scale as a screening tool. *Health and Quality of Life Outcomes*, *8*, 8. https://doi.org/10.1186/1477-7525-8-8.

Hamilton, M. (1959). The assessment of anxiety states by rating. *British Journal of Medical Psychology*, *32*, 50–55. https://doi.org/10.1111/j.2044-8341.1959.tb00467.x.

Hinz, A., Klein, A. M., Braehler, E., Glaesmer, H., Luck, T., Riedel-Heller, S. G., Wirkner, K., & Hilbert, A. (2017). Psychometric evaluation of the generalized anxiety disorder screener GAD-7, based on a large german general population sample. *Journal of Affective Disorders*, *210*, 338–344. https://doi.org/10.1016/j.jad.2016.12.012.

Johnson, S. U., Ulvenes, P. G., Øktedalen, T., & Hoffart, A. (2019). Psychometric Properties of the General anxiety disorder 7-Item (GAD-7) scale in a Heterogeneous Psychiatric Sample. *Frontiers in Psychology*, *10*, 1713. https://doi.org/10.3389/fpsyg.2019.01713.

Jordan, P., Shedden-Mora, M. C., & Löwe, B. (2017). Psychometric analysis of the generalized anxiety disorder scale (GAD-7) in primary care using modern item response theory. *PloS One*, *12*(8), e0182162. https://doi.org/10.1371/journal.pone.0182162.

Kertz, S., Bigda-Peyton, J., & Bjorgvinsson, T. (2013). Validity of the generalized anxiety Disorder-7 scale in an Acute Psychiatric Sample. *Clinical Psychology & Psychotherapy*, *20*(5), 456–464. https://doi.org/10.1002/cpp.1802.

Kline, R. B. (2010). *Principles and practice of structural equation modeling*. Guilford Press.

Konkan, R., Senormanci, O., Guclu, O., Aydin, E., & Sungur, M. Z. (2013). Validity and reliability study for the turkish adaptation of the generalized anxiety Disorder-7 (GAD-7)scale. *Noropsikiyatri Arsivi-Archives of Neuropsychiatry*, *50*(1), 53–58. https://doi.org/10.4274/npa.y6308.

Kroenke, K., Spitzer, R. L., Williams, J. B. W., Monahan, P. O., & Löwe, B. (2007). Anxiety Disorders in Primary Care: Prevalence, impairment, Comorbidity, and detection. *Annals of Internal Medicine*, *146*(5), 317–325. https://doi.org/10.7326/0003-4819-146-5-200703060-00004.

Kujanpää, T., Ylisaukko-Oja, T., Jokelainen, J., Hirsikangas, S., Kanste, O., Kyngäs, H., & Timonen, M. (2014). Prevalence of anxiety disorders among finnish primary care high utilizers and validation of finnish translation of GAD-7 and GAD-2 screening tools. *Scandinavian Journal of Primary Health Care*, *32*(2), 78–83. https://doi.org/10.3109/02813432.2014.920597.

Lovibond, S. H., & Lovibond, P. F. (1995). *Manual for the Depression anxiety stress scales* (2nd ed.). Psychology Foundation.

Löwe, B., Decker, O., Müller, S., Brähler, E., Schellberg, D., Herzog, W., & Herzberg, P. Y. (2008). Validation and standardization of the generalized anxiety disorder screener (GAD-7) in the general population. *Medical Care*, *46*(3), 266–274. https://doi.org/10.1097/MLR.0b013e318160d093.

Nyongesa, M. K., Mwangi, P., Koot, H. M., Cuijpers, P., Newton, C. R. J. C., & Abubakar, A. (2020). The reliability, validity and factorial structure of the Swahili version of the 7-item generalized anxiety disorder scale (GAD-7) among adults living with HIV from Kilifi, Kenya. *Annals of General Psychiatry*, *19*(1), 62. https://doi.org/10.1186/s12991-020-00312-4.

Patriquin, M. A., & Mathew, S. J. (2017). The neurobiological mechanisms of generalized anxiety disorder and chronic stress. *Chronic Stress*, *1*, 2470547017703993. https://doi.org/10.1177/2470547017703993.

Putnick, D. L., & Bornstein, M. H. (2016). Measurement Invariance Conventions and reporting: The state of the art and future directions for Psychological Research. *Developmental Review: DR*, *41*, 71–90. https://doi.org/10.1016/j.dr.2016.06.004.

Sawaya, H., Atoui, M., Hamadeh, A., Zeinoun, P., & Nahas, Z. (2016). Adaptation and initial validation of the Patient Health Questionnaire-9 (PHQ-9) and the generalized anxiety Disorder-7 questionnaire (GAD-7) in an arabic speaking lebanese psychiatric outpatient sample. *Psychiatry Research*, *239*, 245–252. https://doi.org/10.1016/j.psychres.2016.03.030.

Sousa, T. V., Viveiros, V., Chai, M. V., Vicente, F. L., Jesus, G., Carnot, M. J., Gordo, A. C., & Ferreira, P. L. (2015). Reliability and validity of the portuguese version of the generalized anxiety disorder (GAD-7) scale. *Health and Quality of Life Outcomes*, *13*, 50. https://doi.org/10.1186/s12955-015-0244-2.

Spielberger, C. D., Gorsuch, R. L., Vagg, P. R., & Jacobs, G. A. (1983). *Manual for the state-trait anxiety inventory*. Consulting Psychologists Press.

Spitzer, R. L., Kroenke, K., Williams, J. B. W., & Löwe, B. (2006). A brief measure for assessing generalized anxiety disorder: The GAD-7. *Archives of Internal Medicine*, *166*(10), 1092–1097. https://doi.org/10.1001/archinte.166.10.1092.

Tong, X., An, D., McGonigal, A., Park, S. P., & Zhou, D. (2016). Validation of the generalized anxiety Disorder-7 (GAD-7) among chinese people with epilepsy. *Epilepsy Research*, *120*, 31–36. https://doi.org/10.1016/j.eplepsyres.2015.11.019.

Williams, N. (2014). The GAD-7 questionnaire. *Occupational Medicine*, *64*(3), 224. https://doi.org/10.1093/occmed/kqt161.

Woon, L. S., Hatta, S., & Norlaila, M. (2020). Factor structure of the malay-version generalized anxiety Disorder-7 (GAD-7) questionnaire among patients with diabetes Mellitus. *Medicine and Health-Kuala Lumpur*, *15*(1), 208–217. https://doi.org/10.17576/MH.2020.1501.19.

Zigmond, A. S., & Snaith, R. P. (1983). The hospital anxiety and depression scale. *Acta Psychiatrica Scandinavica*, *67*(6), 361–370. https://doi.org/10.1111/j.1600-0447.1983.tb09716.x.

Zinchuk, M., Kustov, G., Pashnin, E., Gersamia, A., Rider, F., Yakovlev, A., Voinova, N., Popova, S., Avedisova, A., & Guekht, A. (2021). Validation of the generalized anxiety Disorder-7 (GAD-7) in russian people with epilepsy. *Epilepsy & Behavior*, *123*, 108269. https://doi.org/10.1016/j.yebeh.2021.108269.